

Trixie Pi Zero 2 W – Pigpio Install & Systemd Setup

Step 0 — Prerequisites

Make sure your system is up-to-date:

```
sudo apt update
sudo apt upgrade -y
sudo apt install build-essential python3-dev python3-venv git -y
```

Step 1 — Download and build pigpio

```
cd ~
git clone https://github.com/joan2937/pigpio.git
cd pigpio
make
sudo make install
```

Check: pigpiod should exist:

```
ls -l /usr/local/bin/pigpiod
```

Step 2 — Fix library path

```
echo "/usr/local/lib" | sudo tee /etc/ld.so.conf.d/usrlocal.conf
sudo ldconfig
```

Check:

```
ldd /usr/local/bin/pigpiod
```

- `libpigpio.so.1` should be found at `/usr/local/lib/libpigpio.so.1`.
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Step 3 — Test the daemon manually

1. Make sure no old daemon is running:

```
sudo killall pigpiod
sudo rm -f /var/run/pigpio.pid
```

2. Start daemon manually:

```
sudo /usr/local/bin/pigpiod
```

3. Check that it's running:

```
ps aux | grep pigpiod
```

- You should see `/usr/local/bin/pigpiod` running.
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Step 4 — Install Python module

Inside your Python environment (or globally):

```
cd ~/pigpio
python3 setup.py install
```

Test Python connection (don't skip):

```
import pigpio
pi = pigpio.pi()
print(pi.connected) # Should print True
pi.stop()
```

Step 5 — Create a systemd service

```
sudo nano /etc/systemd/system/pigpiod.service
```

Paste:

```
[Unit]
Description=Pigpio daemon
After=network.target

[Service]
Type=forking
ExecStartPre=/bin/rm -f /var/run/pigpio.pid
ExecStart=/usr/local/bin/pigpiod
PIDFile=/var/run/pigpio.pid
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

Save and exit.

Step 6 — Enable and start service

```
sudo systemctl daemon-reload
sudo systemctl enable pigpiod
sudo systemctl start pigpiod
sudo systemctl status pigpiod
```

- You should see: `Active: active (running)`
- The daemon will now start automatically on boot.

Step 7 — Final Python test

```
import pigpio
pi = pigpio.pi()
print(pi.connected) # True
pi.stop()
```

 If this works, everything is fully operational.

This **avoids the old issues we ran into**:

- No distutils / Python install errors
 - No `libpigpio.so` not found
 - No invalid `-n` or `-g` options for `pigpiod`
 - PID file issues handled automatically
 - Systemd works reliably on boot
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